

CS 372 Project 3 Traceroute

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Summary:

This program is an extension of the ping skeleton code that was provided, and an implementation of traceroute using some of the preexisting ping functions. The traceroute program begins by printing the current hop number and the final destination the echo request packets hope to reach. For each packet, a TTL, RTT, Type and Code of the reply to the echo request, in addition to the description of the error or information message, before the router that the hop reaches, are printed to the console.

Instructions:

How to run:

- This program can be run with the command
 - `python3 lcmpHelperLibrary.py`
- On my machine, I need to insert the sudo command in front of the execution command
 - `sudo python3 lcmpHelperLibrary.py`

Screenshots:

Host 1: gaia.cs.umass.edu

Start of Program

```
Keegans-MBP:CS372_Project3_Traceroute keeganriley$ sudo python3 lcmpHelperLibrary.py
Password:
Hop: 1:-----Sending 3 packets-----
Pinging (gaia.cs.umass.edu) 128.119.245.12
  TTL=1    RTT=4 ms    Type=11    Code=0    (Time to Live Exceeded in Transit)  192.168.1.1
  TTL=1    RTT=1 ms    Type=11    Code=0    (Time to Live Exceeded in Transit)  192.168.1.1
  TTL=1    RTT=1 ms    Type=11    Code=0    (Time to Live Exceeded in Transit)  192.168.1.1
Hop: 2:-----Sending 3 packets-----
  TTL=2    RTT=15 ms   Type=11    Code=0    (Time to Live Exceeded in Transit)  142.255.32.1
  TTL=2    RTT=15 ms   Type=11    Code=0    (Time to Live Exceeded in Transit)  142.255.32.1
  TTL=2    RTT=20 ms   Type=11    Code=0    (Time to Live Exceeded in Transit)  142.255.32.1
Hop: 3:-----Sending 3 packets-----
  TTL=3    RTT=19 ms   Type=11    Code=0    (Time to Live Exceeded in Transit)  68.173.200.234
  TTL=3    RTT=19 ms   Type=11    Code=0    (Time to Live Exceeded in Transit)  68.173.200.234
  TTL=3    RTT=28 ms   Type=11    Code=0    (Time to Live Exceeded in Transit)  68.173.200.234
Hop: 4:-----Sending 3 packets-----
  TTL=4    RTT=24 ms   Type=11    Code=0    (Time to Live Exceeded in Transit)  68.173.198.50
  TTL=4    RTT=23 ms   Type=11    Code=0    (Time to Live Exceeded in Transit)  68.173.198.50
  TTL=4    RTT=16 ms   Type=11    Code=0    (Time to Live Exceeded in Transit)  68.173.198.50
Hop: 5:-----Sending 3 packets-----
  TTL=5    RTT=18 ms   Type=11    Code=0    (Time to Live Exceeded in Transit)  107.14.19.22
  TTL=5    RTT=34 ms   Type=11    Code=0    (Time to Live Exceeded in Transit)  107.14.19.22
  TTL=5    RTT=25 ms   Type=11    Code=0    (Time to Live Exceeded in Transit)  107.14.19.22
```

Request timeouts. I have explained below my reasoning for leaving the Request timed out and Request timed out (By Exception) below.

```
Hop: 8:-----Sending 3 packets-----
*      *      *      *      *      Request timed out.
*      *      *      *      *      Request timed out (By Exception).
*      *      *      *      *      Request timed out.
*      *      *      *      *      Request timed out (By Exception).
*      *      *      *      *      Request timed out.
*      *      *      *      *      Request timed out (By Exception).
Hop: 9:-----Sending 3 packets-----
TTL=9   RTT=39 ms   Type=11   Code=0   (Time to Live Exceeded in Transit)  198.71.47.58
TTL=9   RTT=24 ms   Type=11   Code=0   (Time to Live Exceeded in Transit)  198.71.47.58
TTL=9   RTT=25 ms   Type=11   Code=0   (Time to Live Exceeded in Transit)  198.71.47.58
```

Program completing running, with ping statistics for the final packets.

```
Hop: 17:-----Sending 3 packets-----
TTL=17   RTT=31 ms   Type=11   Code=0   (Time to Live Exceeded in Transit)  128.119.0.121
TTL=17   RTT=47 ms   Type=11   Code=0   (Time to Live Exceeded in Transit)  128.119.0.121
TTL=17   RTT=29 ms   Type=11   Code=0   (Time to Live Exceeded in Transit)  128.119.0.121
Hop: 18:-----Sending 3 packets-----
TTL=18   RTT=31 ms   Type=11   Code=0   (Time to Live Exceeded in Transit)  128.119.3.33
TTL=18   RTT=34 ms   Type=11   Code=0   (Time to Live Exceeded in Transit)  128.119.3.33
TTL=18   RTT=31 ms   Type=11   Code=0   (Time to Live Exceeded in Transit)  128.119.3.33
Hop: 19:-----Sending 3 packets-----
TTL=19   RTT=26 ms   Type=11   Code=0   (Time to Live Exceeded in Transit)  128.119.3.32
TTL=19   RTT=26 ms   Type=11   Code=0   (Time to Live Exceeded in Transit)  128.119.3.32
TTL=19   RTT=25 ms   Type=11   Code=0   (Time to Live Exceeded in Transit)  128.119.3.32
Hop: 20:-----Sending 3 packets-----
TTL=20   RTT=31 ms   Type=11   Code=0   (Time to Live Exceeded in Transit)  128.119.240.253
TTL=20   RTT=36 ms   Type=11   Code=0   (Time to Live Exceeded in Transit)  128.119.240.253
TTL=20   RTT=29 ms   Type=11   Code=0   (Time to Live Exceeded in Transit)  128.119.240.253
Hop: 21:-----Sending 3 packets-----
ICMP Reply Packet for sequence number: 1 is Valid
TTL=21   RTT=361509 ms   Type=0   Code=0   Identifier=30972   Sequence Number=1   128.119.245.12
ICMP Reply Packet for sequence number: 2 is Valid
TTL=21   RTT=361535 ms   Type=0   Code=0   Identifier=30972   Sequence Number=2   128.119.245.12
ICMP Reply Packet for sequence number: 3 is Valid
TTL=21   RTT=361574 ms   Type=0   Code=0   Identifier=30972   Sequence Number=3   128.119.245.12
Destination Reached, Echo Reply Returned!
Keegans-MBP:CS372_Project3_Traceroute keeganriley$
```

Host 2: 128.119.245.12 Amherst

Start of program

```
Keegans-MBP:CS372_Project3_Traceroute keeganriley$ sudo python3 IcmpHelperLibrary.py
Password:
Hop: 1:-----Sending 3 packets-----
Pinging (128.119.245.12) 128.119.245.12
  TTL=1    RTT=4 ms    Type=11    Code=0    (Time to Live Exceeded in Transit)  192.168.1.1
  TTL=1    RTT=4 ms    Type=11    Code=0    (Time to Live Exceeded in Transit)  192.168.1.1
  TTL=1    RTT=2 ms    Type=11    Code=0    (Time to Live Exceeded in Transit)  192.168.1.1
Hop: 2:-----Sending 3 packets-----
  TTL=2    RTT=20 ms   Type=11    Code=0    (Time to Live Exceeded in Transit)  142.255.32.1
  TTL=2    RTT=16 ms   Type=11    Code=0    (Time to Live Exceeded in Transit)  142.255.32.1
  TTL=2    RTT=15 ms   Type=11    Code=0    (Time to Live Exceeded in Transit)  142.255.32.1
Hop: 3:-----Sending 3 packets-----
  TTL=3    RTT=37 ms   Type=11    Code=0    (Time to Live Exceeded in Transit)  68.173.200.234
  TTL=3    RTT=21 ms   Type=11    Code=0    (Time to Live Exceeded in Transit)  68.173.200.234
  TTL=3    RTT=16 ms   Type=11    Code=0    (Time to Live Exceeded in Transit)  68.173.200.234
Hop: 4:-----Sending 3 packets-----
  TTL=4    RTT=23 ms   Type=11    Code=0    (Time to Live Exceeded in Transit)  68.173.198.50
  TTL=4    RTT=29 ms   Type=11    Code=0    (Time to Live Exceeded in Transit)  68.173.198.50
  TTL=4    RTT=27 ms   Type=11    Code=0    (Time to Live Exceeded in Transit)  68.173.198.50
Hop: 5:-----Sending 3 packets-----
  TTL=5    RTT=22 ms   Type=11    Code=0    (Time to Live Exceeded in Transit)  107.14.19.22
  TTL=5    RTT=20 ms   Type=11    Code=0    (Time to Live Exceeded in Transit)  107.14.19.22
  TTL=5    RTT=22 ms   Type=11    Code=0    (Time to Live Exceeded in Transit)  107.14.19.22
Hop: 6:-----Sending 3 packets-----
  TTL=6    RTT=34 ms   Type=11    Code=0    (Time to Live Exceeded in Transit)  66.109.5.119
```

Hop 8, 3 packets time out. Hop 10, 2 packets get Type 3, Code 3 (Destination Port Unreachable) errors, 1 packet request timed out. Hop 11 recovers and gets a Type 11, Code 0.

```
Hop: 7:-----Sending 3 packets-----
TTL=7    RTT=19 ms    Type=11    Code=0    (Time to Live Exceeded in Transit)  162.252.69.210
TTL=7    RTT=22 ms    Type=11    Code=0    (Time to Live Exceeded in Transit)  162.252.69.210
TTL=7    RTT=81 ms    Type=11    Code=0    (Time to Live Exceeded in Transit)  162.252.69.210
Hop: 8:-----Sending 3 packets-----
*        *        *        *        *    Request timed out.
*        *        *        *        *    Request timed out (By Exception).
*        *        *        *        *    Request timed out.
*        *        *        *        *    Request timed out (By Exception).
*        *        *        *        *    Request timed out.
*        *        *        *        *    Request timed out (By no remaining time left).
Hop: 9:-----Sending 3 packets-----
TTL=9    RTT=105 ms   Type=11    Code=0    (Time to Live Exceeded in Transit)  198.71.47.58
TTL=9    RTT=23 ms   Type=11    Code=0    (Time to Live Exceeded in Transit)  198.71.47.58
TTL=9    RTT=25 ms   Type=11    Code=0    (Time to Live Exceeded in Transit)  198.71.47.58
Hop: 10:-----Sending 3 packets-----
TTL=10    RTT=471 ms   Type=3     Code=3    (Destination Port Unreachable)  127.0.0.1
TTL=10    RTT=29926 ms Type=3     Code=3    (Destination Port Unreachable)  127.0.0.1
*        *        *        *        *    Request timed out.
*        *        *        *        *    Request timed out (By no remaining time left).
Hop: 11:-----Sending 3 packets-----
TTL=11    RTT=28 ms   Type=11    Code=0    (Time to Live Exceeded in Transit)  18.2.136.90
TTL=11    RTT=37 ms   Type=11    Code=0    (Time to Live Exceeded in Transit)  18.2.136.90
TTL=11    RTT=50 ms   Type=11    Code=0    (Time to Live Exceeded in Transit)  18.2.136.90
```

End of program. Traceroute was successful.

```
Hop: 17:-----Sending 3 packets-----
  TTL=17  RTT=29 ms   Type=11   Code=0   (Time to Live Exceeded in Transit)  128.119.0.121
  TTL=17  RTT=31 ms   Type=11   Code=0   (Time to Live Exceeded in Transit)  128.119.0.121
  TTL=17  RTT=37 ms   Type=11   Code=0   (Time to Live Exceeded in Transit)  128.119.0.121
Hop: 18:-----Sending 3 packets-----
  TTL=18  RTT=30 ms   Type=11   Code=0   (Time to Live Exceeded in Transit)  128.119.3.33
  TTL=18  RTT=35 ms   Type=11   Code=0   (Time to Live Exceeded in Transit)  128.119.3.33
  TTL=18  RTT=30 ms   Type=11   Code=0   (Time to Live Exceeded in Transit)  128.119.3.33
Hop: 19:-----Sending 3 packets-----
  TTL=19  RTT=27 ms   Type=11   Code=0   (Time to Live Exceeded in Transit)  128.119.3.32
  TTL=19  RTT=23 ms   Type=11   Code=0   (Time to Live Exceeded in Transit)  128.119.3.32
  TTL=19  RTT=27 ms   Type=11   Code=0   (Time to Live Exceeded in Transit)  128.119.3.32
Hop: 20:-----Sending 3 packets-----
  TTL=20  RTT=39 ms   Type=11   Code=0   (Time to Live Exceeded in Transit)  128.119.240.253
  TTL=20  RTT=25 ms   Type=11   Code=0   (Time to Live Exceeded in Transit)  128.119.240.253
  TTL=20  RTT=30 ms   Type=11   Code=0   (Time to Live Exceeded in Transit)  128.119.240.253
Hop: 21:-----Sending 3 packets-----
ICMP Reply Packet for sequence number: 1 is Valid
  TTL=21  RTT=242575 ms   Type=0   Code=0   Identifier=39509   Sequence Number=1   128.119.245.12
ICMP Reply Packet for sequence number: 2 is Valid
  TTL=21  RTT=242596 ms   Type=0   Code=0   Identifier=39509   Sequence Number=2   128.119.245.12
ICMP Reply Packet for sequence number: 3 is Valid
  TTL=21  RTT=242623 ms   Type=0   Code=0   Identifier=39509   Sequence Number=3   128.119.245.12
Destination Reached, Echo Reply Returned!
Keegans-MBP:CS372_Project3_Traceroute keeganriley$
```

Host 3: 200.10.227.250 (South America)

Start of program

```
Keegans-MBP:CS372_Project3_Traceroute keeganriley$ sudo python3 IcmpHelperLibrary.py
Password:
Hop: 1:-----Sending 3 packets-----
Pinging (200.10.227.250) 200.10.227.250
  TTL=1    RTT=3 ms    Type=11    Code=0    (Time to Live Exceeded in Transit)  192.168.1.1
  TTL=1    RTT=2 ms    Type=11    Code=0    (Time to Live Exceeded in Transit)  192.168.1.1
  TTL=1    RTT=4 ms    Type=11    Code=0    (Time to Live Exceeded in Transit)  192.168.1.1
Hop: 2:-----Sending 3 packets-----
  TTL=2    RTT=35 ms   Type=11    Code=0    (Time to Live Exceeded in Transit)  142.255.32.1
  TTL=2    RTT=15 ms   Type=11    Code=0    (Time to Live Exceeded in Transit)  142.255.32.1
  TTL=2    RTT=15 ms   Type=11    Code=0    (Time to Live Exceeded in Transit)  142.255.32.1
Hop: 3:-----Sending 3 packets-----
  TTL=3    RTT=15 ms   Type=11    Code=0    (Time to Live Exceeded in Transit)  68.173.200.232
  TTL=3    RTT=16 ms   Type=11    Code=0    (Time to Live Exceeded in Transit)  68.173.200.232
  TTL=3    RTT=22 ms   Type=11    Code=0    (Time to Live Exceeded in Transit)  68.173.200.232
Hop: 4:-----Sending 3 packets-----
  TTL=4    RTT=20 ms   Type=11    Code=0    (Time to Live Exceeded in Transit)  68.173.198.48
  TTL=4    RTT=27 ms   Type=11    Code=0    (Time to Live Exceeded in Transit)  68.173.198.48
```

Packets begin timing out

```
Hop: 6:-----Sending 3 packets-----
  *        *        *        *        *    Request timed out.
  *        *        *        *        *    Request timed out (By Exception).
  *        *        *        *        *    Request timed out.
  *        *        *        *        *    Request timed out (By Exception).
  *        *        *        *        *    Request timed out.
  *        *        *        *        *    Request timed out (By Exception).
Hop: 7:-----Sending 3 packets-----
  *        *        *        *        *    Request timed out.
  *        *        *        *        *    Request timed out (By Exception).
  *        *        *        *        *    Request timed out.
  *        *        *        *        *    Request timed out (By Exception).
  *        *        *        *        *    Request timed out.
  *        *        *        *        *    Request timed out (By Exception).
```

Packets find a router, get a TTL expired, and then get stuck on a “Destination Host Unreachable” Type 3, Code 1 error.

```
Hop: 8:-----Sending 3 packets-----
*      *      *      *      *      Request timed out.
*      *      *      *      *      Request timed out (By Exception).
*      *      *      *      *      Request timed out.
*      *      *      *      *      Request timed out (By Exception).
*      *      *      *      *      Request timed out.
*      *      *      *      *      Request timed out (By Exception).
Hop: 9:-----Sending 3 packets-----
TTL=9   RTT=146 ms   Type=11   Code=0   (Time to Live Exceeded in Transit)  200.229.91.30
TTL=9   RTT=142 ms   Type=11   Code=0   (Time to Live Exceeded in Transit)  200.229.91.30
TTL=9   RTT=149 ms   Type=11   Code=0   (Time to Live Exceeded in Transit)  200.229.91.30
Hop: 10:-----Sending 3 packets-----
TTL=10  RTT=3583 ms  Type=3    Code=1   (Destination Host Unreachable)  200.229.91.30
TTL=10  RTT=3174 ms  Type=3    Code=1   (Destination Host Unreachable)  200.229.91.30
TTL=10  RTT=3173 ms  Type=3    Code=1   (Destination Host Unreachable)  200.229.91.30
Hop: 11:-----Sending 3 packets-----
TTL=11  RTT=3174 ms  Type=3    Code=1   (Destination Host Unreachable)  200.229.91.30
TTL=11  RTT=3174 ms  Type=3    Code=1   (Destination Host Unreachable)  200.229.91.30
```

End of program after max hops (50) expire. Packets between 11 and 47 all returned with Type 3, Code 1 error.

```
Hop: 47:-----Sending 3 packets-----
TTL=47  RTT=3176 ms  Type=3    Code=1   (Destination Host Unreachable)  200.229.91.30
TTL=47  RTT=3172 ms  Type=3    Code=1   (Destination Host Unreachable)  200.229.91.30
TTL=47  RTT=3174 ms  Type=3    Code=1   (Destination Host Unreachable)  200.229.91.30
Hop: 48:-----Sending 3 packets-----
TTL=48  RTT=3174 ms  Type=3    Code=1   (Destination Host Unreachable)  200.229.91.30
TTL=48  RTT=3174 ms  Type=3    Code=1   (Destination Host Unreachable)  200.229.91.30
TTL=48  RTT=3173 ms  Type=3    Code=1   (Destination Host Unreachable)  200.229.91.30
Hop: 49:-----Sending 3 packets-----
TTL=49  RTT=3174 ms  Type=3    Code=1   (Destination Host Unreachable)  200.229.91.30
TTL=49  RTT=3135 ms  Type=3    Code=1   (Destination Host Unreachable)  200.229.91.30
TTL=49  RTT=3213 ms  Type=3    Code=1   (Destination Host Unreachable)  200.229.91.30
Hop: 50:-----Sending 3 packets-----
TTL=50  RTT=3276 ms  Type=3    Code=1   (Destination Host Unreachable)  200.229.91.30
TTL=50  RTT=3174 ms  Type=3    Code=1   (Destination Host Unreachable)  200.229.91.30
TTL=50  RTT=3135 ms  Type=3    Code=1   (Destination Host Unreachable)  200.229.91.30
TTLs Expired
Keegans-MBP:CS372_Project3_Traceroute keeganriley$
```

Host 4: 164.151.129.20 (South Africa)

Start of program

```
Keegans-MBP:CS372_Project3_Traceroute keeganriley$ sudo python3 IcmpHelperLibrary.py
Password:
Hop: 1:-----Sending 3 packets-----
Pinging (164.151.129.20) 164.151.129.20
  TTL=1    RTT=3 ms    Type=11    Code=0    (Time to Live Exceeded in Transit) 192.168.1.1
  TTL=1    RTT=1 ms    Type=11    Code=0    (Time to Live Exceeded in Transit) 192.168.1.1
  TTL=1    RTT=2 ms    Type=11    Code=0    (Time to Live Exceeded in Transit) 192.168.1.1
Hop: 2:-----Sending 3 packets-----
  TTL=2    RTT=41 ms   Type=11    Code=0    (Time to Live Exceeded in Transit) 142.255.32.1
  TTL=2    RTT=32 ms   Type=11    Code=0    (Time to Live Exceeded in Transit) 142.255.32.1
  TTL=2    RTT=15 ms   Type=11    Code=0    (Time to Live Exceeded in Transit) 142.255.32.1
Hop: 3:-----Sending 3 packets-----
  TTL=3    RTT=19 ms   Type=11    Code=0    (Time to Live Exceeded in Transit) 68.173.200.234
  TTL=3    RTT=14 ms   Type=11    Code=0    (Time to Live Exceeded in Transit) 68.173.200.234
  TTL=3    RTT=20 ms   Type=11    Code=0    (Time to Live Exceeded in Transit) 68.173.200.234
Hop: 4:-----Sending 3 packets-----
  TTL=4    RTT=26 ms   Type=11    Code=0    (Time to Live Exceeded in Transit) 68.173.198.50
  TTL=4    RTT=21 ms   Type=11    Code=0    (Time to Live Exceeded in Transit) 68.173.198.50
  TTL=4    RTT=16 ms   Type=11    Code=0    (Time to Live Exceeded in Transit) 68.173.198.50
Hop: 5:-----Sending 3 packets-----
  TTL=5    RTT=18 ms   Type=11    Code=0    (Time to Live Exceeded in Transit) 66.109.6.76
  TTL=5    RTT=19 ms   Type=11    Code=0    (Time to Live Exceeded in Transit) 66.109.6.76
  TTL=5    RTT=42 ms   Type=11    Code=0    (Time to Live Exceeded in Transit) 66.109.6.76
Hop: 6:-----Sending 3 packets-----
  TTL=6    RTT=30 ms   Type=11    Code=0    (Time to Live Exceeded in Transit) 66.109.5.139
```

Replies begin to time out on Hop 7, but recover on Hop 8

```
Hop: 6:-----Sending 3 packets-----
TTL=6    RTT=30 ms    Type=11    Code=0    (Time to Live Exceeded in Transit)  66.109.5.139
TTL=6    RTT=19 ms    Type=11    Code=0    (Time to Live Exceeded in Transit)  66.109.5.139
TTL=6    RTT=28 ms    Type=11    Code=0    (Time to Live Exceeded in Transit)  66.109.5.139
Hop: 7:-----Sending 3 packets-----
*        *        *        *        *    Request timed out.
*        *        *        *        *    Request timed out (By Exception).
*        *        *        *        *    Request timed out.
*        *        *        *        *    Request timed out (By Exception).
*        *        *        *        *    Request timed out.
*        *        *        *        *    Request timed out (By Exception).
Hop: 8:-----Sending 3 packets-----
TTL=8    RTT=128 ms   Type=11    Code=0    (Time to Live Exceeded in Transit)  171.75.8.81
TTL=8    RTT=94 ms   Type=11    Code=0    (Time to Live Exceeded in Transit)  171.75.8.81
TTL=8    RTT=105 ms  Type=11    Code=0    (Time to Live Exceeded in Transit)  171.75.8.81
Hop: 9:-----Sending 3 packets-----
TTL=9    RTT=91 ms   Type=11    Code=0    (Time to Live Exceeded in Transit)  195.50.124.34
TTL=9    RTT=321 ms  Type=11    Code=0    (Time to Live Exceeded in Transit)  195.50.124.34
TTL=9    RTT=101 ms  Type=11    Code=0    (Time to Live Exceeded in Transit)  195.50.124.34
Hop: 10:-----Sending 3 packets-----
TTL=10   RTT=256 ms   Type=11    Code=0    (Time to Live Exceeded in Transit)  168.209.100.16
TTL=10   RTT=278 ms   Type=11    Code=0    (Time to Live Exceeded in Transit)  168.209.100.16
TTL=10   RTT=266 ms   Type=11    Code=0    (Time to Live Exceeded in Transit)  168.209.100.16
```

Replies time out on Hops 15 and 16, and recover for the final hop (ping), then End of program

```
TTL=14    RTT=265 ms    Type=11    Code=0    (Time to Live Exceeded in Transit)  168.209.131.135
TTL=14    RTT=289 ms    Type=11    Code=0    (Time to Live Exceeded in Transit)  168.209.131.135
Hop: 15:-----Sending 3 packets-----
*        *        *        *        *    Request timed out.
*        *        *        *        *    Request timed out (By Exception).
*        *        *        *        *    Request timed out.
*        *        *        *        *    Request timed out (By Exception).
*        *        *        *        *    Request timed out.
*        *        *        *        *    Request timed out (By Exception).
Hop: 16:-----Sending 3 packets-----
*        *        *        *        *    Request timed out.
*        *        *        *        *    Request timed out (By Exception).
*        *        *        *        *    Request timed out.
*        *        *        *        *    Request timed out (By Exception).
*        *        *        *        *    Request timed out.
*        *        *        *        *    Request timed out (By Exception).
Hop: 17:-----Sending 3 packets-----
ICMP Reply Packet for sequence number: 1 is Valid
  TTL=17    RTT=546012 ms    Type=0    Code=0    Identifier=39349    Sequence Number=1    164.151.129.20
ICMP Reply Packet for sequence number: 2 is Valid
  TTL=17    RTT=546284 ms    Type=0    Code=0    Identifier=39349    Sequence Number=2    164.151.129.20
ICMP Reply Packet for sequence number: 3 is Valid
  TTL=17    RTT=546535 ms    Type=0    Code=0    Identifier=39349    Sequence Number=3    164.151.129.20
Destination Reached, Echo Reply Returned!
Keegans-MBP:CS372_Project3_Traceroute keeganriley$
```

Comments and Questions:

Validating and Printing:

I implemented the second objective (print expected and actual value to console in case of an error) using `__validateIcmpReplyPacketWithOriginalPingData` and some getters and setters for validating the sequence number, identifier, and raw data. If they are identical, I set the “isValid” attributes to True, and if not, I pass the expected value into an “expectedDict” dictionary. The dictionary is passed back to `__validateIcmpReplyPacketWithOriginalPingData`, and then passed to `printResultToConsole` where the expected and actual values are printed if there is a discrepancy, or not if the packets are valid. This [Ed Discussion #132](#), post informed my decision of how to validate the data and pass it to `printResultToConsole`.

Using numpy to calculate ping statistics:

I included numpy to quickly get the statistics for the RTT calculations. [Answer to Ed Discussion #162](#). I included a private data member in `IcmpPacket` called `__rtt`, which will hold the Round Trip Time of each packet sent. I included a getter and setter method for `__rtt` and only pass it to the `rttList` variable in the `__sendIcmpEchoRequest` if the RTT is valid.

To check if a packet has been sent and received successfully, I have added two private data members to the `IcmpPacket` class, `__isSent` and `__responseReceived`. I have included getters and setters for both to check if a packet is sent and the echo reply is received. I then “get” these two values in the `__sendIcmpEchoRequest` to calculate the packet loss.

I am only marking a packet as received, if it has been validated in the `__validateIcmpReplyPacketWithOriginalPingData`. In this case, an echo reply packet with invalid data (a different sequence number, identifier, or raw data than the echo request sent packet) as a “dropped packet.” Thus, any packet that is not valid, contributes to the packet loss statistic printed to the console.

Inclusion of print info for 3 traceroute echo requests:

I chose to leave the TTL, RTT, Type, Code, error message, and IP address of the router for all three traceroute packets. I’m not sure if I missed an instruction somewhere, but it seems like some of the example outputs on the Ed Discussion only show one RTT for each hop. I left in all three and included the other information for a reason expressed in this video ([Traceroute \(tracert\) Explained - Network Troubleshooting](#)). If one packet gets lost or has an exaggerated RTT, the other packets can show this detail. See below for a screenshot of a hop where the first 2 packets’ requests timed out, and the third returned with a Type 3, Code 0: “Destination Network Unreachable” error.

```
Hop: 19-----
*      *      *      *      *      Request timed out.
*      *      *      *      *      Request timed out (By Exception).
*      *      *      *      *      Request timed out.
*      *      *      *      *      Request timed out (By Exception).
TTL=19   RTT=26653 ms   Type=3   Code=0   (Destination Network Unreachable)  162.125.15.44
```

Duplicate “Request timed out” print statements:

There are two “Request timed out” print statements (the second being “Request Timed out (By Exception)”) that seemed to print every time together. This is because of the “try” and “except” block in `sendEchoRequest`. The first statement signals a timeout, and the second is the timeout actually occurring. I was originally going to clean up the console by removing one and just printing that a timeout had occurred. However, while running the IP address 164.151.129.20, I found that, while there were many duplicates, there were also some that triggered a third statement, “Request timed out (By no remaining time left).” I left all statements in to differentiate the `sendEchoRequest` behaviors. See below for an example of this behavior.

```
Hop: 41-----
*      *      *      *      *      Request timed out.
*      *      *      *      *      Request timed out (By Exception).
*      *      *      *      *      Request timed out.
*      *      *      *      *      Request timed out (By no remaining time left).
TTL=41   RTT=3 ms   Type=3   Code=3   (Destination Port Unreachable)  127.0.0.1
```

I also found this behavior in this example from [Ed Discussion #153](#).

Reported Error Types and Codes:

I have chosen to implement the following ICMP Error Types and Codes in my program. I am including the unreachable errors, as those seem to be the most likely in this scenario. Because we are provided with the IP addresses, and they don’t seem to be deceptive, I am choosing to

forego the “unknown” errors (Type 3, Codes 6 and 7). I have also included TTL expired (Type 11, Code 0) as this is essential for the Traceroute portion of the program. I also left out Source Quench (Type 4, Code 0), which was included in the modules, because it is labeled “Deprecated” on the [IANA website](#). I have also left off “bad IP header” (Type 12, Code 0), since our program is providing that. I included “Fragment Reassembly Time Exceeded” to differentiate a successful traceroute return with a true error.

Type	Code	Description
0	0	echo reply
3	0	destination network unreachable
3	1	destination host unreachable
3	2	destination protocol unreachable
3	3	destination port unreachable
8	0	echo request
11	0	TTL expired
11	1	Fragment Reassembly Time Exceeded

Maximum number of Hops:

In [Ed Discussion #153](#), a max hop limit of 50 is recommended. Our TTL is initially set to 255 for each `IcmpPacket` object. I decided to implement the 50 hop limit in a for loop. This means that the program will stop 1) if the Type 0, Code 0 Echo Reply is returned and all three packets made the trip 2) if the max hops (50) aka the TTL upper limit expires.